Graph-level supervision, not node-level self-supervision, improves low-label molecular property prediction under scaffold splits

Nguyen Ngoc Dung

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Abstract

Self-supervised pretraining is widely reported to improve molecular property prediction where labelled data are scarce, yet reported gains are frequently small relative to the variation introduced by seed and split selection. Whether such pretraining confers a genuine advantage therefore depends on the rigour of the evaluation applied to it. We evaluate node-level and graph-level pretraining on the Tox21 toxicity benchmark under a protocol fixed before any measurement: Bemis–Murcko scaffold splits that place unseen molecular frameworks in the test partition, pretraining corpora decontaminated against every held-out scaffold, repeated seeds at each level of supervision, and decision thresholds specified in advance. A complete two-factor design separates the contribution of each pretraining level and allows their interaction to be estimated rather than assumed. Node-level self-supervision yields no improvement distinguishable from seed variation, whereas graph-level supervised pretraining improves prediction at every level of supervision examined, with the largest advantage where labelled data are most limited. The two levels contribute additively, and no synergy between them is detectable. Substituting a random split for a scaffold split inflates the same benchmark by more than any pretraining effect this study measures, indicating that the evaluation protocol governs the conclusion more strongly than the choice of pretraining objective. Code, split manifests and per-run metrics records are available at: `https://github.com/ngocdung03/mol-ssl`

1 Introduction

Labelled molecular property data are costly to obtain, and pretraining on unlabelled or auxiliary corpora is the standard response. Reported gains are frequently modest relative to the variation induced by seed and split choice, which makes the evaluation protocol a determinant of the reported effect rather than a neutral background.

Hu et al. [1] distinguish two levels at which a graph neural network may be pretrained. Node-level objectives, such as attribute masking, regularise individual atom representations; graph-level objectives fit a supervised multi-task head on pooled graph representations. Their central observation is that strategies operating at only one level yield limited improvement and can produce negative transfer, whereas the two levels applied in sequence perform best. Their reported averages across eight MoleculeNet datasets are 67.0 ROC-AUC without pretraining, 71.4 for node-level attribute masking, 68.9 for graph-level supervision alone, and 73.5 for the two combined.

The present study asks whether that ordering reproduces on a single dataset under an evaluation protocol designed to make small effects distinguishable from noise, and separates the contribution of each level using a complete two-factor design.

2 Methods

2.1 Evaluation protocol

All splits are Bemis–Murcko scaffold splits at proportions 0.8/0.1/0.1, generated once and stored as a manifest that the training code loads rather than recomputes. The Tox21 dataset comprises 7,823 molecules across twelve assay tasks, partitioned into 6,258 training, 782 validation and 783 test molecules. The split is deterministic and independent of seed, so the seed varies only the labelled subsample and the model initialisation. Random splits are not available to the training path; they are confined to a separate comparison module used solely for the inflation measurement in Section 3.1.

Model selection uses the validation partition. The test partition is evaluated once, at the end of training, using the validation-selected parameters. The reported epoch is the early-stopping epoch and never the epoch of best test performance. Every reported quantity is a mean and sample standard deviation over the five seeds $\{0, 1, 2, 3, 4\}$.

2.2 Model and objectives

The encoder is a five-layer GINE network with 300 hidden units, batch normalisation, dropout 0.1, and mean pooling as the readout. The prediction head is a single linear layer over the pooled representation, with twelve outputs for Tox21. The loss masks unobserved labels; a missing assay is not treated as a negative.

Two pretraining objectives were implemented behind a common interface. Node-level attribute masking zeroes the atom-type block of a randomly selected 15% of atoms and reconstructs the hidden element from the surrounding graph context. Graph-level supervision fits a linear head on the pooled graph representation against 128 PCBA bioassays under the same masked loss. Only encoder parameters transfer to the downstream task; both pretraining heads are discarded.

2.3 Factorial design

Node-level and graph-level pretraining were treated as two binary factors, giving four conditions that differ only in the encoder parameters at the start of fine-tuning:

	graph-level absent	graph-level present
node-level absent	`none`: random initialisation	`graph`: PCBA supervision
node-level present	`node`: attribute masking	`node_graph`: masking, then PCBA

The combined condition applies the two stages in the order specified by Hu et al.: node-level self-supervision first, then graph-level supervision initialised from the resulting encoder. Splits, seeds, labelled subsamples, optimiser and learning rate are identical across all four conditions. A single fixed fine-tuning learning rate was used for every condition; tuning one condition and not another is a recognised route to a spurious difference.

2.4 Corpus decontamination

Node-level pretraining used ZINC250k. Of 249,455 molecules, 10,335 (4.14%) shared a Bemis–Murcko scaffold with a downstream validation or test partition and were removed, leaving 239,120. Graph-level pretraining used PCBA. Of 437,927 molecules, 18,809 (4.30%) were removed on the same criterion, leaving 419,118. Filtering was applied against the held-out scaffolds of all four downstream manifests, comprising 1,565 scaffolds for Tox21 and 3,116 in total.

The filter is re-applied at the start of every pretraining run rather than trusted from the cached corpus, and a run whose corpus fails the check is refused. A negative control in which ten Tox21 validation and test molecules were inserted into a corpus was rejected with a non-zero exit status and no checkpoint written. The regression test for the filter is verified by mutation: disabling the filter body causes the test to fail, so a test that cannot fail is not accepted as evidence.

2.5 Pre-registered analysis

Hypotheses and thresholds were recorded before the sweep was executed. The following ordering was predicted from Hu et al.:

$$\texttt{node\_graph} > \texttt{node} > \texttt{graph} > \texttt{none}.$$

A single primary comparison was nominated in advance: `node_graph` against `none` at full supervision. For the remaining fifteen comparisons, a condition is declared to exceed noise only when $|\Delta| > 2\sigma_{\text{pooled}}$, where σ_{pooled} is the root sum of squares of the two seed standard deviations. The unadjusted criterion $|\Delta| > \sigma_{\text{pooled}}$ is reported alongside for continuity with earlier work on the same benchmark. The interaction contrast

$$\Delta_{\text{int}} = (\texttt{node\_graph} - \texttt{graph}) - (\texttt{node} - \texttt{none}) = \texttt{node\_graph} - (\texttt{graph} + \texttt{node}) + \texttt{none} \quad (1)$$

was specified as the test of whether the two levels combine additively. This contrast is introduced in the present study; the two-factor layout follows Hu et al., whose analysis is comparative and does not evaluate an interaction.

3 Results

3.1 Split-induced inflation

An identical model trained on identical data, differing only in the partitioning rule, attained 0.7383 ± 0.0070 AUROC under a scaffold split and 0.8226 ± 0.0146 under a random split. The inflation is 0.0843 ± 0.0140 AUROC and was positive on all five seeds. This magnitude exceeds every pretraining effect reported below and establishes the scale against which those effects should be read.

3.2 The node by graph factorial

Both pretraining stages converged. Node-level masking loss decreased from 0.1518 to 0.0338 over twenty epochs on 239,120 molecules. Graph-level assay loss decreased to 0.0364 from random initialisation and to 0.0352 when initialised from the node-level encoder. The three pretrained encoders were confirmed pairwise distinct, so no two conditions represent the same parameters under different labels.

Table 1 and Figure 1 report the four conditions. The combined condition attained the highest value at every labelled fraction. Table 2 reports the corresponding differences from random initialisation.

Three observations follow. First, node-level pretraining alone satisfied the adjusted criterion at none of the five labelled fractions, and at a labelled fraction of 25% its difference was negative. Second, graph-level pretraining satisfied the unadjusted criterion at all five labelled fractions and the adjusted criterion at three. Third, the primary pre-specified comparison, `node_graph`

Table 1: Test AUROC on Tox21 under a Bemis–Murcko scaffold split, mean and sample standard deviation over five seeds. Run `0903_1750_tox21_2x2_full`; four conditions at five labelled fractions, 100 fine-tuning runs.

Labelled fraction	n_{train}	`none`	`node`	`graph`	`node_graph`
5%	313	0.6215 ± 0.0442	0.6368 ± 0.0277	0.6745 ± 0.0208	0.6936 ± 0.0307
10%	626	0.6253 ± 0.0291	0.6631 ± 0.0316	0.6970 ± 0.0196	0.7168 ± 0.0118
25%	1,564	0.6937 ± 0.0114	0.6879 ± 0.0181	0.7229 ± 0.0080	0.7300 ± 0.0138
50%	3,129	0.7083 ± 0.0198	0.7194 ± 0.0073	0.7502 ± 0.0074	0.7554 ± 0.0120
100%	6,258	0.7293 ± 0.0063	0.7505 ± 0.0106	0.7571 ± 0.0120	0.7676 ± 0.0179

Table 2: Differences in AUROC from random initialisation. Marks denote the pre-registered criteria: $*$ exceeds σ_{pooled}; $**$ exceeds $2\sigma_{\text{pooled}}$; unmarked entries lie inside seed noise.

Labelled fraction	`node`	`graph`	`node_graph`
5%	$+0.0154$	$+0.0530^{*}$	$+0.0722^{*}$
10%	$+0.0377$	$+0.0716^{**}$	$+0.0914^{**}$
25%	-0.0058	$+0.0292^{**}$	$+0.0363^{**}$
50%	$+0.0110$	$+0.0419^{*}$	$+0.0470^{**}$
100%	$+0.0212^{*}$	$+0.0278^{**}$	$+0.0384^{**}$
Fractions exceeding σ_{pooled}	1 of 5	5 of 5	5 of 5
Fractions exceeding $2\sigma_{\text{pooled}}$	0 of 5	3 of 5	4 of 5

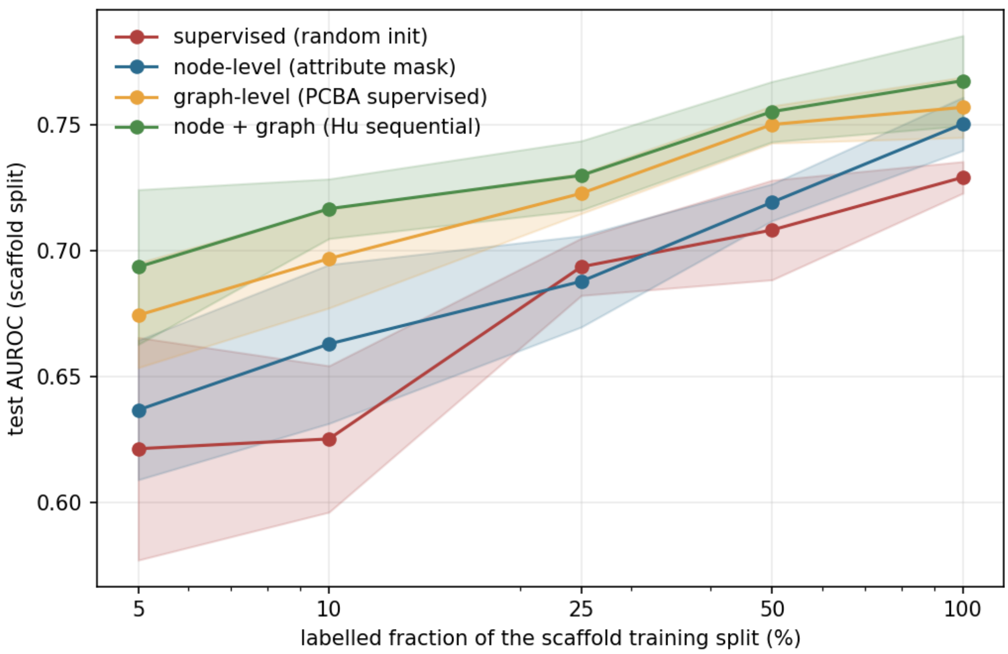

Figure 1: Test AUROC on Tox21 against the labelled fraction of the scaffold training split, for the four pretraining conditions. Bands span one sample standard deviation.

against `none` at full supervision, gave +0.0384 with $\sigma_{\text{pooled}} = 0.0190$ and satisfied the adjusted criterion; being nominated in advance, it requires no multiplicity adjustment.

The advantage of the combined condition increased as supervision was reduced, from +0.0384 at full supervision to +0.0722 at a labelled fraction of 5%. The `node` column was measured five weeks after an independent sweep of the same condition and agreed with it within σ_{pooled} at every labelled fraction, the largest discrepancy being 0.0096.

3.3 Interaction

The contrast of Equation 1 took the values +0.0038, −0.0179, +0.0129, −0.0059 and −0.0107 at the five labelled fractions, against pooled standard deviations of 0.0640, 0.0486, 0.0267, 0.0254 and 0.0248. Every value lies inside seed noise and the sign alternates. The two objectives therefore contribute additively, with no detectable synergy.

Applying the same contrast to the averages of Hu et al. gives $73.5-(68.9+71.4)+67.0 = +0.2$, also consistent with additivity. Their result rests on two positive main effects that accumulate, rather than on an interaction between them. The present finding is therefore consistent with their data on this point.

4 Discussion

The ordering of the two main effects is reversed relative to Hu et al. In their averages, graph-level supervision alone was nearly inert, at 68.9 against a baseline of 67.0, with negative transfer on two of eight datasets, and node-level masking carried most of the improvement. In the present study, graph-level supervision accounted for the larger part of the effect and node-level masking contributed little.

Several differences between the two studies could account for this reversal, and all are hypotheses rather than results. Scale is the most probable: Hu et al. pretrained at the node level on two million ZINC15 molecules, against 239,120 after decontamination in the present study, leaving the node-level encoder correspondingly less well trained. The graph-level corpora differ in kind, their ChEMBL panel spanning 1,310 broad biochemical assays against the 128 bioactivity screens of PCBA, which plausibly resemble the twelve Tox21 endpoints more closely and so transfer more effectively. Their figures are means over eight datasets, from which one dataset may depart without contradicting them; attribute masking produced negative transfer on ToxCast in their own results. Decontamination was asymmetric, their Section 5.1 filtering only the graph-level corpus while both corpora were filtered in the present study. The encoders also differ, theirs a GIN with integer atom-type embeddings against a GINE with continuous atom features and per-layer edge features.

A second observation concerns the interpretation of published gains. A standard pretraining corpus contained 4.30% of molecules sharing a scaffold with the downstream evaluation partitions. A pipeline that omits this filter pretrains on evaluation chemotypes, and the resulting difference cannot be attributed to transfer. Combined with the inflation of 0.0843 AUROC obtained by substituting a random split, this indicates that protocol choices span a range wider than the effects under study.

Limitations

The graph-level corpus is PCBA, comprising 437,927 molecules and 128 assays, in place of the ChEMBL corpus of Hu et al., which contains approximately 456,000 molecules and 1,310 assays. Molecule counts are comparable; task breadth is approximately one tenth. The node-level corpus is smaller by a larger margin: 239,120 decontaminated ZINC250k molecules against the two million ZINC15 molecules of Hu et al. The node-level condition is therefore under-trained relative to theirs, and the absence of a measurable node-level effect in this study should not be read as evidence that the objective is ineffective at scale. Evaluation is restricted to Tox21, so the generality of the reversed main effects is untested, and the proposed resemblance between the graph-level corpus and the downstream endpoints cannot be examined. One node-level objective was implemented; context prediction, edge prediction and contrastive objectives were not. Pretraining ran for twenty epochs at each stage, a schedule that was not tuned. The threshold at $2\sigma_{\text{pooled}}$ is a pragmatic guard against multiplicity and not a hypothesis test; paired per-seed tests or a mixed model over seeds and labelled fractions would be preferable. The corpus was filtered against all four downstream manifests rather than Tox21 alone, which is mildly unfavourable to the graph-level conditions.

References

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